

Sanjiv Motilal Choudhari

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PROFESSIONAL SUMMARY

Data Scientist with a software engineering foundation and hands-on experience building machine learning and AI solutions across predictive modeling, NLP, and graph-based analytics. Proficient in end-to-end data workflows using Python and SQL, spanning data preparation, feature engineering, model development, and deployment. Experienced applying data-driven approaches to real-world problems through scalable, production-ready AI solutions.

EXPERIENCE

Data Analyst (Application Processor)

September 2024 – December 2025

Northeastern University

Boston, MA

- Built scalable SQL and Python (Pandas) data validation pipelines to identify duplicate, missing, and inconsistent applicant records across 30,000+ applications, implementing rule-based quality checks that improved data accuracy and reduced downstream processing errors by 25%.
- Automated the evaluation of international academic credentials by developing Python-based GPA normalization logic and reference mappings, reducing manual validation effort by 40% while improving consistency across diverse grading systems.
- Partnered with Quality Control and admissions teams to analyze data quality issues within Slate CRM, defining validation rules, auditing large applicant datasets, and improving the reliability and efficiency of admissions decision workflows.

Senior Software Engineer

August 2021 – July 2023

Capgemini Technology Services

Chennai, India

- Developed and enhanced property and casualty insurance policy administration features within Guidewire PolicyCenter for Horace Mann, implementing business rules for underwriting, policy validation, and premium calculations while integrating REST APIs to support enterprise insurance workflows serving thousands of users.
- Wrote SQL queries to validate policy data, investigate production issues, and support application debugging, while designing Selenium WebDriver automation frameworks in Java that improved regression testing efficiency and reduced post-release production defects by 35%.
- Performed root cause analysis of production defects by debugging application logic and validating business workflows, reducing manual intervention in underwriting processes by 30%.

PROJECTS

Automated AI Critique and Peer Review

February 2025 – April 2025

github.com/SanjivDS/CS6120_NLP_FinalProject

- Developed an AI-assisted academic peer review system using BART for abstractive summarization and a fine-tuned Mistral 7B model to generate structured manuscript critiques, streamlining the review process while maintaining high-quality feedback.
- Integrated plagiarism detection, bias analysis, and evaluation modules into a unified NLP pipeline, achieving 90% agreement with expert reviewers and demonstrating the effectiveness of LLM-assisted scientific review.

Offside Detection and Diagnostic Analysis github.com/SkillCorner/opencvdata

September 2025 – December 2025

- Developed a Graph Neural Network (GNN) framework to model player interactions from SkillCorner spatiotemporal tracking data, representing players as dynamic graph nodes and ball possession as contextual features to automate offside detection with 85% precision.
- Designed a diagnostic analysis pipeline to evaluate prediction errors across game scenarios, optimizing graph construction and temporal feature representation to improve model robustness for real-world football analytics.

SKILLS

Machine Learning & AI: Scikit-learn, TensorFlow, Keras, PyTorch, Pandas, NumPy, SciPy, Feature Engineering, Model Evaluation, Deep Learning, Time Series Forecasting, Graph Neural Networks (GNNs), Natural Language Processing (NLP), Large Language Models (LLMs), Transformer Models, Prompt Engineering, Model Fine-tuning

Programming Languages: Python, SQL, R, Java, Gosu, C

Tools: Tableau, Power BI, Matplotlib, Streamlit, MySQL, MongoDB, REST APIs, Git, Selenium WebDriver, JIRA

EDUCATION

Northeastern University

Boston, MA

Master of Science, Data Science. GPA: 3.76 / 4.0.

December 2025

Nitte Meenakshi Institute of Technology

Bengaluru, India

Bachelor of Engineering in Computer Science and Engineering. GPA: 8.85 / 10.0.

July 2021